

Fig. 4: Actual-size PCB of the solar light for a portable toilet

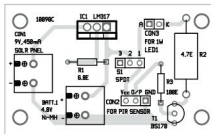


Fig. 5: Component layout of the PCB

a suitable small box in such a way that it can be fitted in the toilet.

After assembling the PCB board, connect the fully-charged battery

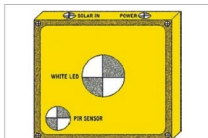


Fig. 6: Suggested enclosure

pack BATT.1 to the circuit and switch on S1. The PIR motion sensor detector requires an initialisation time of about 60 seconds (or even up to three minutes). Thereafter, it enters standby mode, ready for detection.

You can adjust the delay pot-meter in PIR sensor clockwise to increase the time delay from three seconds to up to 300 seconds. Similarly, turning the sensitivity pot in PIR sensor clockwise increases the detection sensitivity from three

to six metres. In some PIR sensor potmeters, the LDR socket option is not provided.

The solar panel (not fitted within the enclosure) can be mounted at an appropriate location. Battery pack BATT.1 is best linked to the circuit via two flexible, well-insulated multi-strand wires. Then, a heat-sink should be added to IC LM317.

Finally, house the assembled PCB in a small plastic case with holes drilled for the connectors for linking the solar panel, switch S1, LED1 and PIR motion sensor. The suggested enclosure is shown in Fig. 6.

EFY note. Connect LED1 externally, with a proper heat-sink, to CON3 in the PCB. **EFY**



T.K. Hareendran is an electronics hobbyist, freelance technical writer and circuit designer

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